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Frau Dr. Pooja Gupta
Department of Stem Cell Biology
Universitätsklinikum Erlangen
Kussmaulallee 4
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Postdoctoral Researcher – AI-based Multi-Omics in Computational Biomedicine

Dear Dr. Gupta,

Three years of experience translating continuous biological models into high-performance Python architectures have prepared me to build the AI frameworks required for your BMFTR-funded project, "AI-Driven Multi-Omics Integration for Predicting IBD-PD Comorbidity Progression". Your group's focus on utilizing machine learning to decode the gut-brain axis and comorbidity networks targets exactly the intersection of complex high-dimensional systems and deep learning that I am passionate about. My transition from computational systems modeling to bioinformatics pipeline engineering makes me a highly suitable candidate to build the AI frameworks your research requires.

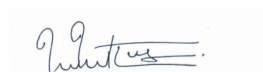
While my academic background is in systems modeling, I have spent the last year intensively retraining in modern bioinformatics and machine learning. At my current internship with HealthTwist GmbH, I am solely responsible for refactoring a production medical data pipeline (in R/tidyverse) for Germany's interventional radiology quality registry. Working with 143,000 patient records and 1,920 variables, I externalized hardcoded logic into configuration files, verified byte-level reproducibility, and reduced the 1,834-line codebase by 26%, dramatically improving maintainability.

Your project's need for graph neural networks, variational autoencoders, and integrative molecular interaction networks aligns closely with my extensive experience analyzing coupled systems. I bring a strong foundation in the tools you require:

- **AI / Deep Learning Foundations:** During my post-doc, I led the translation of a legacy Fortran model into a high-performance Python engine using **Google JAX**. By engineering branchless logic and vectorized multidimensional array operations, I achieved direct compatibility with **Google TPUs and GPUs** for massive parallelization. As JAX shares core paradigms with **PyTorch** and **TensorFlow**, this provided the foundational expertise required to scale the graph neural networks and autoencoders your project demands.
- **Pipeline Development & ML Workflows:** Actively building ML workflows for tabular clinical data using Tidymodels (R) and expanding my deep learning capabilities via IBM's "Machine Learning with Python" and "Introduction to AI" training. Highly proficient in building reproducible workflows (Nextflow) and processing multi-omic data.
- **Large-Scale Data & High-Performance Computing:** Processed terabytes of raw experimental data during my Ph.D. and developed algorithms for HPC clusters. My expertise navigating complex, integrated networks and running simulation-based perturbation analyses aligns perfectly with modeling IBD-PD comorbidity progression.

I am deeply motivated by the Universitätsklinikum Erlangen's mission to leverage AI for biomarker discovery and translational impact in complex diseases. My robust mathematical background combined with modern data engineering skills will allow me to immediately contribute to your computational infrastructure. I welcome the opportunity to discuss further and am available for an interview.

Sincerely



Dr. Thejus Mahajan